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THE NATIONAL COLLEGE BASAVANAGUDI, BENGALURU- 560 004

AUTONOMOUS

NAAC Accredited 'A' Grade

Affiliated to the Bengaluru City University

Website: www.ncbgudi.com

Curriculum

Structure

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Bachelor of Computer

Applications -BCA

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B.Sc in Computer Science

**STATE EDUCATION POLICY – 2024 (SEP-
2024)**

Effective from Academic Year 2024-25

BCA Course Matrix – JULY -2025

B C A Semester III - Year 2025-26

	Subjects			Marks			
Code	Course	L	P	Exam	IA	Total	Credits
24BCA31	Database Management System	4	-	80	20	100	3
24BCA32	Python Programming	4	-	80	20	100	3
24BCA33	Computer Networks	4	-	80	20	100	3
24BCA34	Cyber Security	2		40	10	50	2
24BCA31P	Database Management System Lab	-	3	40	10	50	2
24BCA32P	Python Programming Lab	-	3	40	10	50	2
24BCA33P	Computer Networks Lab	-	3	40	10	50	2
24BCAL31	Language 1	4	-	80	20	100	3
24BCAL32	Language 2	4	-	80	20	100	3
		22	09	560	140	700	23

B C A Semester IV - Year 2025-26

	Subjects			Marks			
Code	Course	L	P	Exam	IA	Total	Credits
24BCA41	Artificial Intelligence	4	-	80	20	100	3
24BCA42	Design and Analysis of Algorithms	4	-	80	20	100	3
24BCA43	Internet Technologies	4	-	80	20	100	3
24BCA44	Ethical Hacking	2	-	40	10	50	2
24BCA41P	Artificial Intelligence Lab	-	3	40	10	50	2
24BCA42P	Design and Analysis of Algorithms Lab	-	3	40	10	50	2
24BCA43P	Internet Technologies Lab	-	3	40	10	50	2
24BCASE1	Probability and Statistics	2	-	40	10	50	2
24BCAL41	Language 1	4	-	80	20	100	3
24BCAL42	Language 2	4		80	20	100	3
		24	09	600	150	750	25

Course Code : 24BCA31			
TITLE : Database Management Systems			
Lecture Hrs. 56	Internal Marks : 20	Exam Marks : 80	Credits: 3
Course Outcomes (Cos): 1. CO1: Understand the fundamental concepts of data, database systems, DBMS architecture, data models and the various types of DBMS along with their classifications. 2. CO2: Design high-level conceptual data models using the Entity-Relationship approach, apply design principles and understand physical storage structures, file organization, and indexing mechanisms. 3. CO3: Apply the relational model concepts, perform normalization using functional dependencies and write effective SQL queries to manage and manipulate data constraints and views. 4. CO4: Utilize relational algebra to query databases, analyze query processing and optimization techniques and explain transaction management, concurrency control and recovery mechanisms in DBMS. Basics of PL/SQL.			
Unit-I: Fundamentals of Database Systems and Architecture			14 Hours
Introduction to Data and Database, History of Database. Characteristics of the Database Approach. Significance and Advantages of Database Management Systems. Actors on the Scene, Workers behind the Scene. System Structure: Instance and Schema, Data Models, Data Independence. Three Schema Architecture. Database Languages and Interfaces. The Database System Environment. Centralized and Client-Server Architecture. Classification of Database Management System.			
Unit-II: Database Design and Storage Structures			14 Hours
High-Level Conceptual Data Models for Database Design. Entity Types, Entity Sets, Attributes and Keys. Relationship Types, Relationship Sets, Roles and Structural Constraints. Weak Entity Types. Extended ER Features. Refining the ER Design. Naming Conventions and Design Issues. ER to Relational Mapping. File Organization and Storage. Secondary Storage Devices. File Organization Techniques. Single-Level Ordered Index. Multi-Level Indexes. Indexes on Multiple Keys			
Unit-III: Relational Model,	14		

Normalization and SQL	Hours
Relational Model Concepts. Relational Model Constraints and Relational Database Schema. Update Operations and Dealing with Constraint Violations. Anomalies in a Database. Functional Dependency. Armstrong's Axioms. Closure of a Relation and Attributes. Lossless Join and Dependency Preservation. Normalization: 1NF, 2NF, 3NF, BCNF. Structure of Relational Databases. SQL: Data Definition and Data Types. Specifying Constraints in SQL. Schema Change Statements. Insert, Delete and Update Statements. Views (Virtual Tables). Assertions and Triggers.	
Unit-IV: Query Processing, Transactions and PL/SQL	14 Hours
Unary Relational Operations: SELECT and PROJECT. Relational Algebra Operations. Binary Relational Operations: JOIN and DIVISION. Additional Relational Operations. Query Processing and Optimization: Evaluation of Relational Algebra Expressions. Query Equivalence. Introduction to Transaction Processing. Transaction and System Concepts. States of a Transaction. Desirable Properties of Transactions (ACID). Concurrency Control Techniques: Two-Phase Locking Techniques. Backup and Recovery from Failures, Basics of PL/SQL Programming.	
Text Books:	
1. Elmasri R. and Navathe S.B., <i>Fundamentals of Database Systems</i>, 7th Edition, Addison-Wesley, 2016. 2. Silberschatz A., Korth H.F., and Sudarshan S., <i>Database System Concepts</i>, 7th Edition, Tata	

McGraw Hill, 2019.

3. Ivan Bayross, *SQL, PL/SQL – The Programming Language of Oracle*, 4th Edition, BPB Publications.

Reference:

1. C.J. Date, A. Kannan, S. Swamynathan, *An Introduction to Database Systems*, 8th Edition, Pearson Education, 2009.
2. Raghu Ramakrishnan and Johannes Gehrke, *Database Management Systems*, 3rd Edition, McGraw Hill, 2003.
3. Kevin Loney and George Koch, *Oracle 9i – The Complete Reference*, McGraw-Hill International Edition.

Course Code : 24BCA32			
TITLE : Python Programming			
Lecture Hrs. 56	Internal Marks : 20	Exam Marks : 80	Credits: 3
Course Description: Python Programming introduces the fundamentals of coding using Python, a powerful and beginner-friendly language. The course covers variables, control structures, functions, data structures, OOP, and file handling.			
Course Outcomes (Cos): CO1: Understand the basic concepts and apply them to develop programs. CO2: Demonstrate proficiency in handling Python’s complex data structures and apply file handling for various formats. CO3: Apply object-oriented programming concepts in Python and use libraries for efficient data manipulation and analysis. CO4: Utilize Python packages and APIs for effective data visualization and apply them in real-time data analysis projects.			
Unit-I: Foundations of Python Programming			14 Hours
Introduction to Python: Python Interpreter/Shell, Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Type hint, Indentation, Comments, Reading Input, Print Output, Type Conversions, The type() function and is operator, Dynamic and Strongly Typed Language. Control Flow: Conditional blocks: if, else, elif, Nested if. Match case, Looping: while, for, range, loop manipulation using break, continue, else, pass. Functions: Function Definition and Calling, Built-In Functions, Return Statement, Default Parameters, Scope and Lifetime of Variables, Command Line Arguments. Strings: Creating and Storing Strings, String Operations, Slicing, Joining, String Methods.			
Unit-II: Data Structures and File Handling			14 Hours
Lists: Creating Lists, Basic List Operations, Indexing and Slicing, List Methods, The del Statement. Dictionaries: Creating Dictionaries, Accessing and Modifying Key-Value Pairs, Dictionary Methods. Tuples: Creating Tuples, Tuple Operations, Indexing, Slicing, Tuple Methods, Relationships between Tuples, Lists, and Dictionaries. Sets: Creating Sets, Set Operations, Set Methods. Iterators and Iterables. File Handling: Types of Files, Reading and Writing Text and Binary Files, CSV File Handling.			

Unit-III: Object-Oriented Programming and Data Handling Libraries	14 Hours
Object-Oriented Programming: Classes and Objects in Python, Constructor Method, Multiple Objects, Class vs Data Attributes, Encapsulation, Inheritance, Polymorphism. Introduction to Python Libraries for Data Handling: NumPy – Arrays and Operations, Pandas – Series and DataFrames, Indexing and Querying, Handling Missing Values, Data Aggregation, Grouping, and Summarization.	
Unit-IV: Data Analysis and Visualization	14 Hours
Importing and Exporting Data (CSV, JSON), Understanding and Formatting Data. Using Matplotlib and Plotly for Visualization, Generating and Plotting Data (Line Graphs, Bar Charts), Working with APIs, Introduction to Seaborn: Distribution Plots (distplot, histplot), Categorical Plots (barplot, countplot), Styling and Aesthetics.	
Text Books:	
1. Wesley J. Chun, <i>Core Python Applications Programming</i>, 3rd Edition, Pearson Education, 2016. 2. Yashavant Kanetkar and Aditya Kanetkar, <i>Let Us Python</i>, 3rd Edition, BPB Publications. 3. Jeeva Jose & P. Sojan Lal, <i>Introduction to Computing and Problem Solving with Python</i>, Khanna Publishers, New Delhi, 2016. 4. Eric Matthes, <i>Python Crash Course – A Hands-On, Project-Based Introduction to Programming</i>, 2nd Edition, No Starch Press, 2019. 5. Gowrishankar S, Veena A, <i>Introduction to Python Programming</i>, 1st Edition, CRC Press/Taylor & Francis, 2018. ISBN: 978-0815394372.	
Reference:	
1. Allen B. Downey et al., <i>How to Think Like a Computer Scientist: Learning with Python</i>, John Wiley, 2015. 2. John Zelle, <i>Python Programming: An Introduction to Computer Science</i>, 2nd Edition, Course Technology, Cengage Learning, 2013. 3. A.N. Kamthane & A.A. Kamthane, <i>Programming and Problem Solving with Python</i>, McGraw Hill Education, 2017. 4. Mark Lutz, <i>Learning Python</i>, 5th Edition, O'Reilly Publications, 2013. ISBN: 9781449355739. 5. Ljubomir Perkovic, <i>Introduction to Computing Using Python – An Application Development Focus</i>, Wiley, 2012.	

Course Code : 24BCA33			
TITLE : Computer Networks			
Lecture Hrs. 56	Internal Marks : 20	Exam Marks : 80	Credits: 3
Course Description: Computer Networks explores the concepts of data communication, network architectures, and protocols. The course covers OSI & TCP/IP models, IP addressing, routing, switching, and network security.			
Course Outcomes (Cos): CO1: Understand the basics of data communication CO2: Explore data link layer protocols and understand channelization methods CO3 Analyse the functioning of the network layer, routing algorithms and protocols CO4: Comprehend transport layer protocols and application layer services			
Unit-I: Introduction to Networking and Communication Principles			14 Hours
Introduction: Data Communications, Networks, Network Types, Internet History, Network Models: Protocol Layering, The OSI Model, TCP/IP Protocol Suite, Introduction to Physical Layer: The types of Transmission media, Transmission Impairments, Data Rate Limits, Performance.			

Unit-II: Data Link Layer and MAC Techniques	14 Hours
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Introduction to Data Link Layer: Link-Layer Addressing. Error Detection and Correction: Block Coding, Cyclic Codes, Checksum. Data Link Control: Data-Link Layer Protocols, HDLC, Point-To-Point (PPP), MAC sublayer, Logical Link Control Layer, Random access protocols: Pure ALOHA, Slotted ALOHA, CSMA, CSMA/CD, CSMA/CA, Scheduling approaches to medium access control: Reservation systems, Polling, Token Passing rings, Comparison of Random access and Scheduling access control, Comparison of Random access and scheduling medium access control, Channelization: FDMA, TDMA, CDMA.

Unit-III: Network Layer Fundamentals and Routing Protocols	14 Hours
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Introduction to Network Layer: Network-Layer Services, Packet Switching, Network-Layer Performance, IPV4 Addresses, Network Layer Protocols: Internet Protocol (IP), ICMPv4, Mobile IP, Unicast Routing: Routing Algorithms, Unicast Routing Protocols, Next Generation IP: IPv6 Addressing.

Unit-IV: Transport and Application Layer Protocols with QoS Mechanisms	14 Hours
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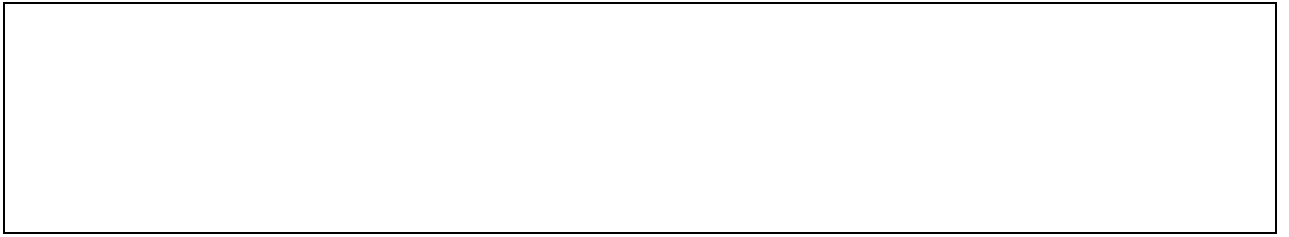
Introduction to Transport Layer: Transport-Layer Protocols, User Datagram Protocol, Transmission Control Protocol: TCP Services, TCP Features, Segment, TCP Connection, TCP Congestion Control, Flow Control, Error Control, Application Layer: WWW, E-MAIL, Domain Name System (DNS), Quality of Service: Flow Control To Improves QoS, Integrated Services.

Text Books:

1. Behrouz A. Forouzan, “Data Communications and Networking”, 5th Edition, McGraw Hill Education,2013.

Reference:

1. Andrew S. Tanenbaum, David J. Wetherall, “Computer Networks”, 5th Edition, Prentice Hall,2011.
2. Larry L. Peterson and Bruce S. Davie, “Computer Networks A System Approach”, 5th Edition,MKP, 2012.
3. James F. Kurose , Keith W. Ross, “ Computer Networking, A Top-Down Approach”, 5thEdition, Pearson, 2012



Course Code : 24BCA34			
TITLE : Cyber Security			
Lecture Hrs. 28	Internal Marks : 10	Exam Marks : 40	Credits: 2
Course Description: Cyber Security introduces the principles of securing digital systems, networks, and data from unauthorized access and threats. The course covers cryptography, network security, ethical hacking, malware analysis, and security policies.			
Course Outcomes (Cos): CO1: Understand the fundamentals of cyber security, types of cyber threats and adopt safe internet practices and responsible digital behavior. CO2: Understand the basic concepts of cryptography, encryption techniques and password security and secure communication protocols used for data protection. CO3: Apply system and personal security measures to protect devices and data from threats, vulnerabilities and online attacks.			
Unit-I: Fundamentals of Cyber Security & Secure Communication			14 Hours
Introduction to Cyber Security: Importance of cyber safety in the digital world, Common types of cyber threats: viruses, worms, trojans, phishing, ransomware, fake websites, social engineering, Real-life examples of cyber attacks. Goals of Cyber Security: Confidentiality, Integrity, Availability (CIA Triad),Difference between threats, vulnerabilities, and risks. Security Concepts and Terminology: Firewall, antivirus, malware, hacking. Safe Internet Practices: Responsible digital behaviour, Secure browsing and identifying fake/unsafe websites, Email safety: detecting phishing and spam, Use of public Wi-Fi and avoiding data leaks. Introduction to Cryptography and Secure Communication: Purpose of encryption and decryption, Difference between symmetric and asymmetric encryption, Role of HTTPS and SSL, Two-Factor Authentication (2FA), OTP-based logins, Importance of secure messaging and email			
Unit-II: System, Personal, and Device Security Practices			14 Hours

System and Device Security Basics: Operating System security basics, User authentication and access control, Password policies and best practices, Antivirus and anti-malware software. Web and App Security: Common web security issues: SQL injection, XSS, CSRF (overview only), Mobile app security – permissions and safe usage, Safe use of mobile devices, laptops, and social media, Software updates and patch management. Network and Online Protection: Introduction to firewalls and VPNs, Risks of pop-ups, cookies, and online tracking, Safe and responsible data sharing online	
Text Books:	
1. Chirag Shah (2018). A Hands-On Introduction to Cybersecurity. Wiley.	
Reference:	
1. Nina Godbole & Sunit Belapure (2011). Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives. Wiley India. 2. Rao, U. & Nayak, P. (2014). Cyber Security. Cengage Learning. Moeti J. (2021). Cybersecurity for Beginners. Amazon Digital Services (for simple reading-level coverage).	

Course Code : 24BCA31P	
TITLE : DBMS Lab	
Total No. of Teaching Hours: 42	No. of Hours per Week : 3

Internal Marks : 10					Exam Marks : 40		Credits: 2		
Course Description: DBMS Lab provides hands-on experience in designing and managing relational databases using SQL. Students practice creating schemas, querying data, and implementing constraints, joins, and transactions.									
Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)									
Course Outcomes (COs) CO1 CO2 CO3 CO4 CO5 CO6 CO7 CO8	Program Outcomes (POs)								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
	Note: Course Articulation Matrix relates course outcomes of course with the corresponding program outcomes whose attainment is attempted in this course. Mark =X ‘in the intersection cell if a course outcome addresses a particular program outcome.								
List of Practical:									
1. Create a table STUDENT with the following fields: RollNo, Name, DOB, Department, Marks. Insert at least 5 records Display all records Update marks for a specific student Delete a student record									
2. Create a table COURSE with CourseID, CourseName, Credits. Alter the table to add a field Department Drop the field Credits									
3. Create two tables EMPLOYEE1 and EMPLOYEE2 with the following attributes: (FNAME, MNAME, LNAME, SSN, BDATE, ADDRESS, SEX, SALARY, SUPERSSN, DNO). Perform UNION, INTERSECT, and MINUS operations on them.									
4. Use the STUDENT table to perform Aggregate functions and Scalar functions									
5. Using the STUDENT and COURSE tables, Find students who scored above average marks using a subquery. List students enrolled in „Computer Science“ using a subquery.									
6. Create DEPARTMENT and STUDENT tables.									

Write queries to display student names along with their department names using JOIN operations.
7. Create a view to show student names and marks from the STUDENT table where Marks >75. Query the view and update the view.
8. Demonstrate the use of GRANT and REVOKE on the STUDENT table. Use COMMIT and ROLLBACK after INSERT and DELETE commands.
Part B
9. Write a PL/SQL block to accept a number and check if it is even or odd.
10. Write a PL/SQL program to divide two numbers and handle the exception if the denominator is zero.
11. Use a parameterized cursor to display students based on department input.
12. Create a stored function to calculate grade based on marks: Above 80: Distinction 60–79: First Class 40–59: Second Class Below 40: Fail
13. Write a stored procedure to update the marks of a student given their RollNo and new marks.
14. Create a BEFORE INSERT trigger on STUDENT to ensure marks are not entered as negative.
15. Create a trigger to log changes into a table STUDENT_LOG whenever the marks are updated.
16. Use a cursor with a loop to count and display the number of students in each department.

Course Code : 24BCA32P		
TITLE : Python Programming Lab		
Total No. of Teaching Hours: 42	No. of Hours per Week : 3	
Internal Marks : 10	Exam Marks : 40	Credits: 2
Course Description: Python Lab offers practical experience in writing and executing Python programs to solve real-world problems. Students learn core concepts like data types, control structures, functions, file handling, and basic object-oriented programming.		

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)

Course Outcomes (COs)	Program Outcomes (POs)								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1									
CO2									
CO3									
CO4									
CO5									
CO6									
CO7									
CO8									

Note: Course Articulation Matrix relates course outcomes of course with the corresponding program outcomes whose attainment is attempted in this course. Mark =X 'in the intersection cell if a course outcome addresses a particular program outcome.

List of Practical:

1. Write a Python program to declare variables, perform arithmetic operations, and display results.
2. Create a program to check if a number is even or odd using if-else.
3. Write a Python program to print the first n Fibonacci numbers using a for loop.
4. Implement a program that accepts a string and counts the number of vowels and consonants.
5. Create a program to store student details in a dictionary and retrieve details based on user input.
6. Demonstrate the use of break, continue, and pass in loops.
7. Write a program to create NumPy arrays, perform element-wise operations, and reshape arrays.
8. Create a Pandas Series and perform indexing, slicing, and querying operations.
9. Load a dataset into a Pandas Data Frame and perform sorting and filtering operations.
10. Write a program to handle missing values by filling them with mean/median values.

Part B

11. String Operations: Write a program to count the occurrences of each word in a given string.
12. Implement a program to insert, delete, and update elements in a list.
13. Create a dictionary with employee details and perform CRUD operations.
14. Implement a program to generate prime numbers up to n using a generator function.
15. Write a Python program to calculate the factorial of a number using recursion.
16. Write a program to read and write student marks into a text file and

display the contents.
17. Plot a line graph and a bar chart using Matplotlib.
18. Write a program to load a CSV file into Pandas and perform basic data analysis.
19. Implement a program to group a dataset by category and calculate summary statistics.
20. Load a dataset from Scikit-learn and display its basic properties.

Course Code : 24BCA33P		Discipline Core 4 Practical	
TITLE: Computer Networks Lab			
Total No. of Teaching Hours: 42		No. of Hours per Week : 4	
Internal Marks : 10	Exam Marks : 40		Credits: 2
Course Objectives/Course Description A lab-based course designed to provide hands-on experience in implementing and understanding core computer networking concepts and protocols.			
Learning Outcome Students will be able to implement, simulate, and troubleshoot basic computer network protocols and understand their real-time applications.			
1.Execute the following commands: arp, ipconfig, hostname, netdiag, netstat,			

- nslookup, pathping, ping route, tracert
- 2. Study of different types of network cables.
- 3. Practically implement the cross-wired cable and straight wired cable using crimping tool.
- 4. Study of network IP address configuration: (Classification of address, static and dynamic address)
- 5. Study of network IP address configuration: (IPv4 and IPv6 , Subnet, Supernet)
- 6. Study of network devices: (Switch, Router, Bridge)
- 7. Configure and Connect the computer in LAN.
- 8. PASSING MESSAGES BETWEEN ROUTERS (USE OF CLOCK AND SERIAL).
- 9. CONFIGURE TELNET.
- 10. CONFIGURE DHCP, DNS AND HTTP SERVER.
- 11. Configuration of wifi hotspot, and connect other devices (mobile / laptop).
- 12. Configuration of switches
- 13. CREATE BUS NETWORK, STAR NETWORK AND HYBRID NETWORK
- 14. CONFIGURATION OF SMART HOME NETWORK.
- 15. Configuration of VPN using Packet Tracer/ GNS3

Course Code : 24BCA41			
TITLE : Artificial Intelligence			
Lecture Hrs. 56	Internal Marks : 20	Exam Marks : 80	Credits: 3
Course Description This course introduces the basic concepts of Artificial Intelligence (AI) in a simple way.Students will learn how machines can think, learn, and make decisions like humans.It also covers real-life applications of AI such as chatbots, robots, and voice assistants			
Course Objectives: To understand the basic concepts, techniques, and real-life applications of Artificial			

Intelligence.	
Course Outcomes (Cos): CO1: Understand the fundamental concepts of Artificial Intelligence and problem-solving strategies. CO2 : Apply logical reasoning techniques, perform inferences, solve constraint satisfaction problems and basic learning methods. CO3: Design AI systems using planning techniques and reasoning, interpret data using perception models. CO4 : Explore the application domains and evaluate ethical and societal implications of AI technologies.	
Unit-I:	14 Hours
Fundamentals of Artificial Intelligence and Search Techniques Definitions, Applications, and Scope. Intelligent Agents: Agents and Environments, Concept of Rationality, Nature of the Environment, Structure of Agents. Knowledge-Based Agents: Introduction, The Wumpus World as an Example. Problem Solving: Problem-Solving Agents, Formulating Problems. Search Techniques: Uninformed Search Strategies – Depth First Search (DFS), Breadth First Search (BFS), Iterative Deepening Search. Informed Search Strategies – Best First Search, A* Search, AO* Search, Means-End Analysis. Adversarial Search and Games: Two-Player Zero-Sum Games, Minmax Algorithm, Alpha-Beta Pruning.	
Unit-II:	14 Hours
Knowledge Representation, Reasoning, and Learning Paradigms Propositional Logic, First-Order Predicate Logic, Differences between Propositional and First-Order Inference. Inference Techniques: Unification and Lifting, Forward Chaining, Backward Chaining, Resolution, Truth Maintenance Systems. Constraint Satisfaction Problems (CSPs): Definition, Examples, Backtracking Search. Learning Concepts: Rote Learning, Learning by Taking Advice, Learning in Problem Solving, Learning from Examples. Decision Trees and Winston's Learning Program.	
Unit-III	14Hours
Planning, Reasoning and Perception Introduction to Planning: Planning Problem, State-Space Search, The Blocks World Problem, STRIPS Representation. Handling Uncertainty: Non-Monotonic Reasoning, Probabilistic Reasoning, Introduction to Fuzzy Logic and Fuzzy Set Theory. Introduction to Perception: Computer Vision – Image Classification, Object Detection. Natural Language Processing (NLP): Introduction, Syntactic Processing, Semantic Analysis, Discourse and Pragmatic Processing.	

Unit-IV:	14 Hours
Machine Learning, Neural Networks, and AI Ethics Types of Learning – Supervised, Unsupervised, and Reinforcement Learning. Neural Networks: Basics of Artificial Neural Networks (ANN), Deep Learning Concepts , Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short Term Memory (LSTM) Networks and their Applications. Expert Systems: Architecture, Components, and Role of Expert Systems with Two Case Studies. Legal and Ethical Issues in AI: Societal Impact, Bias, Privacy, and Accountability in AI Systems..	
Text Books 1. M.C. Trivedi, A Classical Approach to Artificial Intelligence, Khanna Book Publishing Company, 2024 (AICTE Recommended). 2. Nilsson Nils J., Artificial Intelligence: A New Synthesis, Morgan Kaufmann Publishers Inc., ISBN: 978-1-55860-467-4. 3. Dan W. Patterson, Introduction to Artificial Intelligence & Expert Systems, PHI Learning, 2010. 4. Rajiv Chopra, Data Science with Artificial Intelligence, Machine Learning and Deep Learning, Khanna Book Publishing Company, 2024. 5. Russell, S. and Norvig, P., Artificial Intelligence – A Modern Approach, 3rd Edition, Prentice Hall.	
Reference Books 1. M.C. Trivedi, Introduction to AI and Machine Learning, Khanna Book Publishing Company, 2024. 2. Van Hirtum, A. & Kolski, C., Constraint Satisfaction Problems: Algorithms and Applications, Springer, 2020. 3. Elaine Rich, Kevin Knight, Shivasankar B. Nair, Artificial Intelligence, 3rd Edition, McGraw Hill, 2019. 4. Rajiv Chopra, Machine Learning and Machine Intelligence, Khanna Book Publishing Company, 2024..	

Course Code : 24BCA42			
TITLE : Design And Analysis of Algorithms			
Lecture Hrs. 56	Internal Marks : 20	Exam Marks : 80	Credits: 3
Course Description Design and Analysis of Algorithms (ADA) explores the principles behind algorithm development and efficiency analysis. The course covers algorithm design techniques such as divide and conquer, greedy methods, dynamic programming, and graph algorithms, emphasizing time and space complexity.			
Course Objectives: The objective of ADA is to develop the ability to design efficient algorithms using various techniques like divide and conquer, greedy, and dynamic programming. It emphasizes analyzing algorithm performance using time and space complexity. Students learn to apply suitable strategies for solving real-world computational problems.			
Course Outcomes (Cos): CO1: Understand the fundamental principles of algorithm design and analyze algorithm efficiency. CO2 : Apply classical algorithm design paradigms to solve computational problems. CO3: Employ optimization techniques to design efficient solutions for complex problems. CO4 : Analyze the computational complexity of problems and Problem-Solving Methods.			

Unit-I: Fundamentals of Algorithm Design and Computational Efficiency	14 Hours
Introduction: Algorithms, Fundamentals of Algorithmic Problem Solving, Important Problem Types, Fundamental Data Structures. Fundamentals of the Analysis of Algorithm Efficiency: The Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, Mathematical Analysis of Non-recursive and Recursive Algorithms, Empirical Analysis of Algorithms.	
Unit-II: Algorithm Design Techniques	14 Hours
Brute Force Method: Selection Sort and Bubble Sort, Sequential Search, Brute-Force String Matching, Depth-First Search and Breadth-First Search. Decrease and Conquer: Insertion Sort, Topological Sorting, Algorithms for Generating Combinatorial Objects, Decrease by-aConstant-Factor Algorithms. Divide and Conquer: Merge Sort, Quick Sort, Binary Tree Traversals and Related Properties.	
Unit-III: Optimization Techniques	14Hours
Space and Time Tradeoffs: Sorting by Counting, Input Enhancement in String Matching, Hashing. Dynamic programming: Binomial Coefficient, Principle of Optimality, Optimal Binary Search Trees, Knapsack Problem and Memory Functions. Warshall's and Floyd's Algorithms. Greedy Technique: Prim's Algorithm, Kruskal's Algorithm, Dijkstra's Algorithm, Huffman Trees.	
Unit-IV: Understanding Algorithmic Complexity & Problem-Solving Methods	14 Hours
Limitations of Algorithm Power: Lower-Bound Arguments, Decision Trees, P, NP and NP Complete Problems. Limitations of Algorithm Power, Backtracking: 4-Queens problem, Hamiltonian Circuit Problem, Sum of Subset Problem. Branch-and-Bound: Assignment Problem, Knapsack Problem, Traveling Salesman Problem.	
Text Books	
1. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", 3rd Edition, Pearson, 2012. 2. Horowitz, Sahni, Rajasekaran, "Fundamentals of Computer Algorithms", 2/e, Universities Press, 2007.	

Reference Books

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, "Introduction to Algorithms", 3rd Edition, The MIT Press, 2009.
2. A.V. Aho, J.E. Hopcroft, J.D. Ullmann, "The design and analysis of Computer Algorithms", Addison Wesley Boston, 1983.
3. Jon Kleinberg, Eva Tardos, "Algorithm Design", Pearson Education, 2006.

Course Code : 24BCA43			
TITLE : Internet Technologies			
Lecture Hrs. 56	Internal Marks : 20	Exam Marks : 80	Credits: 3

Course Description Internet Technology explores the foundational concepts and technologies that drive the modern internet. The course covers web architecture, HTML, CSS, JavaScript, client-server models, protocols like HTTP, and basics of web application development.	
Course Objectives: It aims to equip learners with the skills to design and develop interactive web pages using HTML, CSS, and JavaScript.	
Course Outcomes (Cos): CO1: Understand the fundamental concepts of the Internet, its services, applications, organizations and apply ethical practices in cyberspace. CO2 : Understand the workings of web technologies and basic concepts of web information retrieval. CO3: Design and develop web applications. CO4 : Work with web application frameworks and databases and explore current research trends.	
Unit-I: Internet Fundamentals, Services and Cyber Ethics	14 Hours
INTERconnected NETwork (Internet): The Giant Wide Area Network, Communicating over the Internet, Accessing the Internet, Internet Organisations, Cyber Ethics, Internet Applications: Internet services, Electronic Mail(E-Mail), File Transfer, Real-Time User Communication, Remote Login, Usenet, World Wide Web: Introduction, Working Web, Web Terminology, Web Architecture, World Wide Web Challenges.	
Unit-II: Web Technologies and Evolution	14 Hours
Hypertext Transfer Protocol (HTTP): HTTP, HTTP Version, HTTP connections, HTTP Communication, Hypertext Transfer Protocol Secure, Hypertext Transfer Protocol State Retention: Cookies, Hypertext Transfer Protocol Cache, Evolution of Web: The Generations of Web, Web 1.0, Web 2.0, Web 3.0, Information Retrieval on the Web (Web IR): Web Information Retrieval, Web Information Retrieval Tools, Web Information Retrieval Architecture (Search Engine Architecture), Web Information Retrieval Performance Metrics, Web Information Retrieval Models.	
Unit-III: Web Development Technologies and Architectures	14Hours

Web Development: Basics of Web Development, Elements of Web Development, Client-Side and Server-Side Scripting, Model-View-Controller Architecture for Web Application Development, Client-Side Technologies: HTML, CSS, JavaScript, Bootstrap Framework, AngularJS Framework, Server-Side Technologies: Server-Side Scripting, Node.js.

Unit-IV: Frameworks and Databases for Web Applications with Research Insights

14 Hours

Web Application Frameworks: Django, Ruby on Rails, Structured Query Language: Relational Databases, Web Databases: NoSQL Databases, Non-relational and Distributed Data, Understanding Popular Databases. Research Trends on the Web: Contextual Information Retrieval, Web Mining.

Text Books

1. Rajkamal, "Internet and Web Technologies", Tata McGraw-Hill Education.
2. Robert W. Sebesta, "Programming the World Wide Web", Pearson, 8th Edition

Reference Books

1. Randy Connolly, Ricardo Hoar, "Fundamentals of Web Development", Pearson
2. Valerie Lampkin, "Modern Web Development with HTML5 and CSS", Cengage Learning.

Course Code:24BCA44			
TITLE: Ethical Hacking			
Lecture Hours: 26	Internal Marks: 10	Exam Marks: 40	Credits: 02
Course Description: This course introduces the fundamental concepts of ethical hacking and cybersecurity. You will learn how to identify system vulnerabilities, perform penetration testing, and implement countermeasures to secure networks and systems. Emphasis is placed on legal and ethical responsibilities while using real-world tools and techniques to simulate cyber-attacks in a controlled environment.			
Course Objectives: CO1: To provide students with a foundational understanding of cybersecurity principles and ethical hacking practices. CO2: To equip students with practical skills to identify security loopholes and perform vulnerability assessments and penetration testing. CO3: To enable students to apply ethical hacking techniques to secure computer systems and networks in real-world scenarios, with a strong emphasis on legal and ethical considerations.			

Course Outcomes (Cos):

COs	Description
CO1	Understand the scope, phases and legal aspects of ethical hacking and recognize various cyber-attacks and the role of ethical hackers in cyber security.
CO1	Apply reconnaissance, scanning and enumeration techniques using tools to gather and analyze system and network information for vulnerability assessment
CO1	Identify and explain common system hacking methods, malware types and social engineering attacks and suggest preventive strategies.
CO1	Understand web and network security threats, secure browsing practices and the use of firewalls, VPNs and IDS/IPS for ethical hacking and protection.

Unit-I Foundations of Ethical Hacking and Cyber Attacks**14 Hours**

Definition and scope of ethical hacking, Difference between hacking and ethical hacking, Types of hackers – white hat, black hat, grey hat. Ethical hacking process and phases – planning, scanning, gaining access, maintaining access, clearing tracks. Common attack types – phishing, malware, password cracking, social engineering attacks such as phishing, baiting, and shoulder surfing. Introduction to attacks including Trojans, worms, viruses, and ransomware. Basics of keyloggers, spyware, backdoors, and rootkits. Introduction to sniffing and packet capturing. Role of ethical hacking in cyber security. Basic prevention strategies and the importance of user awareness.

Unit-II Foot printing, Scanning, Enumeration and System Exploitation**14 Hours**

Introduction to reconnaissance and information gathering. Footprinting techniques – search engines, social networks, WHOIS, DNS, email tracking. Tools for passive and active footprinting. Scanning networks – IP scanning, port scanning, vulnerability scanning. Overview of common tools: Nmap, Angry IP Scanner. Enumeration basics: Windows and Linux enumeration, identifying users, shares, services and open ports, interpreting scan results and identifying risks. Basics of system hacking

including password cracking, privilege escalation, keylogging, and techniques used to gain unauthorized access.

Text Books

1. Bachaalani, P., & Mehta, N. (2020). Fundamentals of Ethical Hacking. BPB Publications.

Reference Books

1. EC-Council (2019). Ethical Hacking and Countermeasures: Attack Phases. Cengage Learning.

2. Kimberly Graves (2010). CEH: Official Certified Ethical Hacker Review Guide. Wiley.

3. Jon Erickson (2008). Hacking: The Art of Exploitation. No Starch Press.

24BCA41P

Title: Artificial Intelligence Lab

Total No. of Teaching Hours: 42

No. of Hours per Week: 03

Internal Marks: 10

Exam Marks: 40

Credits: 2

Course Description: This lab-based course offers hands-on experience in implementing fundamental concepts of Artificial Intelligence (AI). Students will explore key AI techniques such as search algorithms, knowledge representation, inference, and machine learning. Using real-world tools and programming environments, students will build intelligent agents and systems that simulate human decision-making and learning processes.

Course Objectives:

CO1: To introduce students to practical aspects of AI through programming exercises and simulations.

CO2: To implement various AI algorithms such as uninformed and informed search, minimax, and machine learning techniques.

CO3: To develop AI-based solutions for real-world problems using tools such as Python and relevant AI libraries.

CO4: To enhance the ability to design, implement, test, and evaluate intelligent systems and agents.

Course Outcomes (Cos):

CO1: Implement fundamental AI algorithms including search strategies and knowledge-based systems.

CO2: Apply machine learning models to datasets and interpret results.

CO3: Use AI tools and frameworks (such as Python, Scikit-learn, or TensorFlow) to develop intelligent applications.

CO4: Evaluate and compare the performance of AI algorithms in solving specific problems.

CO5: Demonstrate understanding of ethical implications in the design and deployment of AI systems.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos).

Course Outcomes (COs)	Program Outcomes (POs)								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1									
CO2									
CO3									
CO4									
CO5									
CO6									
CO7									

Note: Course Articulation Matrix relates course outcomes of course with the corresponding program outcomes whose attainment is attempted in this course. Mark =X 'in the intersection cell if a course outcome addresses a particular program outcome.

Lab programs
1. Demonstrate basic problem-solving using Breadth-First Search on a simple grid
2. Implement Depth-First Search (DFS) on a small graph.
3. Solve the Water Jug Problem using Breadth First Search (BFS).
4. Implement a Hill Climbing search to find the peak in a numeric dataset.
5. Apply the A* Search algorithm to find the shortest path in a 4x4 grid.
6. Implement the Minmax search algorithm for 2-player games. You may use a game tree with 3 plies.
7. Use constraint propagation to solve a Magic Square puzzle
8. Apply optimization techniques to find the maximum value in a list.
9. Represent and evaluate propositional logic expressions.
10. Implement a basic rule-based expert system for weather classification.
11. Implement a basic AI agent with simple decision-making rules.
12. Implement a basic Rule-Based Chatbot.
13. Using Python NLTK, perform the following Natural Language Processing (NLP) tasks for text content. a) Tokenizing b) Filtering Stop Words c) Stemming d) Part of Speech tagging e) Chunking f) Named Entity Recognition (NER)

24BCA42P		
Title: Design and Analysis of Algorithms Lab		
Teaching Hours: 03 hours/week		No. of Hours per Week: 03
Internal Marks: 10	Exam Marks: 40	Credits:2
Course Description: This lab course provides practical experience in implementing fundamental algorithms using C/C++/Java/Python. It reinforces the concepts learned in algorithm design and analysis theory by guiding students to build, execute, and analyze classical algorithms. Students will explore searching, sorting, divide and conquer, dynamic programming, greedy		

CO3									
CO4									
CO5									
CO6									
CO7									

Note: Course Articulation Matrix relates course outcomes of course with the corresponding program outcomes whose attainment is attempted in this course. Mark =X 'in the intersection cell if a course outcome addresses a particular program outcome.

Lab programs

1 Write a program to implement linear search algorithm. Repeat the experiment for different values of n, the number of elements in the list to be searched and plot a graph of the time taken versus n.

2. Write a program to implement binary search algorithm. Repeat the experiment for different values of n, the number of elements in the list to be searched and plot a graph of the time taken versus n.

3. Write a program to solve towers of Hanoi problem and execute it for different number of disks

4. Write a Program to Sort a given set of numbers using selection sort algorithm. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.

5. Write a program to find minimum and maximum value in a given array using divide and conquer

6. Write a Program to Sort a given set of elements using quick sort algorithm. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n.

7. Write a Program to implement dynamic programming algorithm for the optimal binary search tree.

8. Write a Program to implement Floyd's algorithm and find the lengths of the shortest paths from every pair of vertices in a given weighted graph.

9. Write a Program to solve the string matching problem using Boyer-Moore approach.

10. Write a program to implement BFS traversal algorithm.
11. Write a program to find the minimum spanning tree of a given graph using Prim's algorithm.
12. Write a program to find the minimum spanning tree of a given graph using Kruskal's algorithm.
13. Write a Program to obtain the topological ordering of vertices in a given digraph. Compute the transitive closure of a given directed graph using Warshall's algorithm
14. Write a program to implement backtracking algorithm for solving 4 queens problem.
15. Write a Program to Find a subset of a given set $S = \{s_1, s_2, \dots, s_n\}$ of n positive integers whose sum is equal to a given positive integer d . For example, if $S = \{1, 2, 5, 6, 8\}$ and $d = 9$ there are two solutions $\{1, 2, 6\}$ and $\{1, 8\}$. A suitable message is to be displayed if the given problem instance doesn't have a solution.

24BCA43P		
Title: Internet Technologies Lab		
Teaching Hours: 03 hours/week		No. of Hours per Week: 03
Internal Marks: 10	Exam Marks: 40	Credits:2
Course Description: This lab course introduces students to the fundamentals of internet and web technologies through hands-on exercises. Students will learn how to use basic internet services like email and virtual meetings, build web pages using HTML and CSS, and enhance interactivity using JavaScript. Additionally, the course explores basic server-side scripting using CGI and helps students apply concepts in real-world tasks such as form creation, client-side validation, layout using frames and tables, and building a mini-website.		

CO5									
CO6									
CO7									

Note: Course Articulation Matrix relates course outcomes of course with the corresponding program outcomes whose attainment is attempted in this course. Mark =X 'in the intersection cell if a course outcome addresses a particular program outcome.

Lab programs

1 Demonstrate E-Mail working (Sending ,Receiving, forward).

2. How to create, organize meeting in Zoom/ GoogleMeet.

3. Create a form by using various attributes of the input tags (text box, multiline textbox, option button, check box).

4. Create a simple HTML page by using some of the basic tags (hyperlink, marquee, image).

5. Create a web page with multiple types of style sheet used in a single page.

6. Write a CGI sample program to send output back to the user.

7. Create Time-Table using table tag.

8. Creation of Frames in browser window using HTML.

9. Write a java script program to create dialogue boxes using alert, confirm and prompt methods.

10. Write a java script program on Form Validations.

11. Write a java script program to perform four arithmetic operations: Addition, Subtraction, Multiplication and Division on two numbers.

12. Create a web site of your College.